

LuaPilot Manual

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LuaPilot — User Manual

LuaPilot is a standalone Lua 5.5 binary for Linux scripting and automation. This is the reference manual, organised one file per module so each section stays focused and easy to maintain.

Getting started

- [getting-started.md](#) — install, first script, the three launch modes (folder / embedded / via PATH).
- [security.md](#) — what the hardening does and doesn't protect against, recommended least-privilege patterns.

Modules

Each module lives in its own file under `modules/`:

Module	Purpose
argparse	Command-line argument parser.
exec	Run external programs and capture output.
fs	File system : list, copy, hash, attrs.
http	HTTP client (built on <code>cpp-httplib</code>).
inotify	Filesystem event watching (<code>inotify(7)</code>).
json	JSON encode/decode (<code>nlohmann/json</code>).
logging	Leveled logger.
signal	POSIX signals : <code>SIGTERM</code> , <code>SIGINT</code> , ...
socket	TCP client/server with timeouts.
sqlite	Embedded SQL database.
strings	String manipulation (<code>split</code> , etc.).
sys	<code>which</code> , <code>hostname</code> , <code>env</code> , <code>pid</code> , ...
tables	Table manipulation (<code>mergeTables</code> , <code>deepCopyTable</code>).
time	Monotonic and realtime clocks.
tls	TLS sockets : <code>connect_tls</code> , <code>starttls</code> .
toml	TOML parser (<code>tomlplusplus</code>).
user	System user lookups via NSS.
workers	Parallel jobs (real OS threads).

The [user](#) module is the canonical reference for the documentation pattern : *Why*, *API*, *Quick example*, *Error contract*, *Design decisions*, *Not in v1*. Other modules vary in how strictly they follow that template ; the goal over time is to align them.

Cookbook

[cookbook.md](#) — concrete recipes : directory watch + email, parallel HTTP fetches, graceful shutdown, TOML+SQLite, etc.

Building a PDF

The manual can be exported to a single PDF :

```
cd docs
./build_doc.sh           # uses pandoc + LaTeX
```

See [docs/manual_order_en.txt](#) for the file order ; that's the only place to edit when adding or reordering chapters.

Conventions used in the docs

- **API tables** : every module starts with a small table listing functions, signatures, and what they return.
- **Error contract** : a dedicated section explains which errors are raised (`luaL_error`) versus returned as (`nil`, `err`). This is consistent across all modules.
- **Design decisions** : when a choice is non-obvious (“why not recursive ?”, “why mandatory ?”), there’s a short rationale.
- **Not in v1** : a short list of what could be added additively later without breaking SemVer.

English | [Français](#)

Getting started

Download

Prebuilt binaries for x86_64, aarch64 (RPi4), and armv6l (RPi0) are available on the [releases page](#).

Build from source

LuaPilot vendors all its dependencies (Lua, OpenSSL, SQLite, miniz, nlohmann/json, cpp-httpplib, tomlplusplus), so the only prerequisites on your system are :

- a C++23 compiler (recent GCC or Clang)
- CMake (3.20)
- wget and unzip

```
git clone https://github.com/Chipsterjulien/luapilot_standalone.git
cd luapilot_standalone
./build_local.sh           # downloads deps and compiles
                           # (~5 min on first run, faster afterwards)
./run_tests.sh             # offline harness – should print 827 PASS / 0 FAIL
```

The build script downloads each dependency from its upstream source, verifies the SHA256, then compiles statically. If the upstream is temporarily unreachable ([lua.org](#) notably has had outages), the script falls back to the Internet Archive’s Wayback Machine — the SHA256 check still applies.

The resulting binary is at `test/luapilot`.

Three ways to run a Lua script

LuaPilot supports three launch modes :

1. Folder mode

Point the binary at a directory containing `main.lua` :

```
echo 'print("hello from LuaPilot")' > main.lua
./test/luapilot .
```

`require("X")` in `main.lua` looks for `X.lua` in the same directory.

2. Embedded mode (single-file executable)

Package a script and its modules into a self-contained binary :

```
./test/luapilot --create-exe . myapp
./myapp           # runs main.lua from the embedded ZIP
```

Internally, LuaPilot appends a ZIP to its own binary, and reads `main.lua` (plus any required module) from that ZIP at runtime. The resulting `myapp` is fully self-contained — no LuaPilot install needed on the target machine, just glibc.

3. Embedded via PATH (folder + auto-detect)

If LuaPilot is on \$PATH and you `chmod +x main.lua` after adding a shebang line :

```
#!/usr/bin/env luapilot
print("hello")

chmod +x main.lua
./main.lua
```

LuaPilot uses the script's own directory as the folder, so `require("helpers")` finds `helpers.lua` next to `main.lua`.

Command-line invocation

In addition to the three launch modes above, LuaPilot accepts a few standard flags :

Flag	Effect
-h, --help	Print the help text and exit 0.
-V, --version	Print <code>luapilot <version></code> and exit 0.
-c <dir> <out>, --create-exe <dir> <out>	Create a self-contained executable named <out> by embedding <dir> (which must contain <code>main.lua</code>).

Any other argument starting with - is treated as an **unknown option** : LuaPilot prints `Unknown option: ...` + a hint to use `--help`, and exits 1 instead of trying to interpret it as a directory name. Folders whose name legitimately starts with - can still be passed via `./-dirname` (POSIX convention).

```
luapilot --version    # luapilot 1.7.1
luapilot --help       # full usage
luapilot --bogus      # Unknown option: --bogus
                     # Try 'luapilot --help' for more information.
```

The same version is also exposed to scripts as `luapilot.VERSION` (plus `luapilot.VERSION_MAJOR` / `VERSION_MINOR` / `VERSION_PATCH` as integers) — see [sys](#).

First script

Once the binary is built, try this :

```
-- main.lua
print("LuaPilot says hi")
print("PID:", luapilot.pid())
print("Host:", luapilot.hostname())

local r, err = luapilot.http.get("https://example.com/")
if r then
    print("Status:", r.status)
else
    print("HTTP failed:", err)
end
```

Run it with `./test/luapilot .` (or whichever mode you prefer).

Where to look next

- Each module has its own page under [modules/](#).
- The [user](#) module is a small, complete example of the documentation pattern used throughout — read it as the canonical reference.
- For date and duration utilities (ISO 8601 timestamps, human durations like "5m" or "2h30m"), see [time](#) — the `luapilot.time.*` sub-table covers parsing and formatting.

- See [security.md](#) before exposing LuaPilot scripts to anything that might receive untrusted input.

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Security model

LuaPilot runs trusted Lua scripts. It is not a sandbox.

Scripts execute with the full Lua standard library (`luaL_openlibs`), and LuaPilot additionally exposes filesystem and process primitives such as `exec`, `remove`, `rmdirAll` and `chdir`. A script therefore has the same privileges as the process running it : it can read, modify or delete files and run arbitrary commands. This is by design — like `make`, a shell script, or any build tool, LuaPilot is meant to run code you control.

Only run `main.lua` and `required` modules that you trust. Do **not** use LuaPilot to execute Lua from untrusted sources ; it provides no isolation against hostile code, and none is intended. If you need to run untrusted scripts, use a purpose-built Lua sandbox instead.

What the hardening *does* protect against

The hardening work in LuaPilot protects *legitimate* use against accidents and supply-chain tampering :

- **Path / symlink handling** uses `lexically_relative()` and component-wise `is_within()` guards (never string prefix checks), so a `cp src dst/` cannot escape into a sibling tree via a symlinked path component.
- **Process-group cleanup** in `luapilot.exec` ensures that forked children don't outlive the parent or leak zombie processes when the script is interrupted.
- **Output limits** on `luapilot.exec` bound the amount of stdout and stderr captured into memory (default 16 MiB each, configurable), so a runaway subprocess can't OOM the LuaPilot process.
- **Dependency checksums** : every vendored dependency is SHA256-pinned in `build_local.sh`. An empty hash refuses to build. A mismatch deletes the downloaded file and exits with an error. This protects against a compromised upstream or Wayback Machine archive (which is the fallback source).
- **TLS verification** is on by default (`verify=true`). Disabling it requires an explicit `verify=false` per call — no silent fallback. Custom CA stores can be passed via `ca_cert` or `ca_path`, or via `SSL_CERT_FILE` / `SSL_CERT_DIR` environment variables.

What it does *not* protect against

- A malicious script. LuaPilot has no sandbox. If you `exec` user input, you have a shell injection. If you `loadstring` user input, you have arbitrary code execution.
- A determined attacker who has obtained code execution on the machine running LuaPilot. The hardening makes accidental mistakes loud, not adversarial attacks impossible.
- Long-tail OS-level issues (kernel exploits, container escapes, privilege escalation). LuaPilot is an ordinary userland binary.

Recommended pattern : least privilege

When running LuaPilot as a service, treat it like any other unprivileged process :

As root, create a dedicated service user with no shell.

```
useradd --system \
  --home-dir /var/lib/myapp \
  --create-home \
  --shell /usr/sbin/nologin \
  --comment "myapp LuaPilot service" \
  myapp
```

*# Use `luapilot.user.exists("myapp")` in your install script to
verify the user is provisioned before launching the daemon.*

Combine with systemd unit file options like `User=myapp`, `PrivateTmp=yes`, `ProtectSystem=strict`, `NoNewPrivileges=yes`, and `CapabilityBoundingSet=` (empty unless you genuinely need a capability). The kernel does much more than LuaPilot can to contain a misbehaving script ; let it.

[English](#) | [Français](#)

argparse — command-line argument parser (Lua module)

A declarative argument parser for command-line scripts. Bundled as a pure-Lua module, loaded via `require("argparse")`. Returns results as data (not via prints or `os.exit`) — the script decides what to do.

Why

A LuaPilot script that takes command-line arguments shouldn't have to reinvent argument parsing (positional vs flag, defaults, help text, type coercion, choices) every time. **argparse** makes the common CLI patterns one-liners — and stays out of the script's way by returning everything as data : help is a result, not a side effect ; errors are a return value, not an exception.

API

```
local argparse = require("argparse")
local p = argparse(prog, description?)
```

Builder method	Returns	Use
<code>p:flag(spec, opts?)</code>	<code>p</code> (chainable)	Boolean flag (no value).
<code>p:option(spec, opts?)</code>	<code>p</code> (chainable)	Flag that takes a value.
<code>p:argument(name, opts?)</code>	<code>p</code> (chainable)	Positional argument.
<code>p:parse(src?)</code>	<code>(res, err)</code>	Parse arguments (see below).
<code>p:get_usage()</code>	<code>string</code>	Render the help text manually.

`spec` for `flag` and `option` is a **single string** containing one or more names separated by spaces : `"-v"`, `"--verbose"`, or `"-v --verbose"`. Names must start with `-`.

`name` for `argument` is a plain identifier (no leading `-`).

opts table fields

Field	Type	Effect
<code>help</code>	<code>string</code>	Help text shown by <code>get_usage()</code> .
<code>default</code>	<code>any</code>	Value returned when the flag/option/argument is absent.
<code>dest</code>	<code>string</code>	Force the destination key in <code>res</code> . Default: first long name without dashes, else first short name without dash.
<code>required</code>	<code>boolean</code>	Mark as required. For positionals, defaults to <code>true</code> unless <code>default</code> is set.
<code>choices</code>	<code>array of strings</code>	Restrict the raw value to a finite set.
<code>convert</code>	<code>function string -> any</code>	Coerce the raw string. Returning <code>nil</code> is treated as a failure (the <code>convert</code> function may return <code>(nil, "reason")</code>).

`parse(src?)` return contract

- **Success** : (res, nil) — res is a table indexed by **dest**, containing the parsed value (or default) for each declared flag / option / argument.
- **Help requested** : ({ help = true, usage = "<text>" }, nil) when the user passed -h or --help. Help is a success, not an error. The script decides what to do (typically print res.usage and return).
- **Failure** : (nil, err) — human-readable string for any user input problem (unknown option, missing value, invalid choice, conversion error, missing required argument, extra positional).

`parse()` **never raises** on user input — every user error goes through (nil, err). Only **builder misuse** (empty spec, name without -, duplicate name, etc.) raises via `error()`, because that's a programmer bug.

Source of arguments

- `p:parse()` (no argument) reads the global **arg** table, indices 1..n (stopping at the first hole). Indices ≤ 0 (binary path, folder) are ignored — these are LuaPilot's internal slots, not user arguments.
- `p:parse({ "a", "b" })` parses an explicit array. Useful for testing or for parsing arguments that don't come from the shell.

Accepted argument forms

- `--long val`
- `--long=val`
- `-s val`
- `-s=val`
- `--` ends option parsing — every subsequent token is treated as a positional, even if it starts with -.

Short options collapsed together (`-abc`) and the `-fvalue` form (short option immediately followed by its value, no separator) are **not supported** in v1.

Quick example

```
local argparse = require("argparse")

local p = argparse("myapp", "Process some files.")
:flag("-v --verbose", { help = "Verbose output" })
:option("-o --output",
        { help = "Output file", default = "out.txt" })
:option("-n --count",
        { help = "How many", convert = tonumber, default = 1 })
:argument("input", { help = "Input file" })

local args, err = p:parse()

-- Help is a success, not an error.
if args and args.help then
    print(args.usage)
    return
end

if err then
    io.stderr:write("error: ", err, "\n")
    io.stderr:write(p:get_usage(), "\n")
    os.exit(2)
end

print(args.input, args.output, args.count, args.verbose)
```

Running it :

```
$ ./myapp.lua data.csv --count 5 -v
data.csv    out.txt    5      true
```

```
$ ./myapp.lua --help
Usage: myapp [options] <input>
```

Process some files.

Arguments:

input Input file

Options:

```
-h, --help          show this help
-v, --verbose       Verbose output
-o, --output <value> Output file
-n, --count <value> How many
```

Error contract

- **Builder misuse** (`p:flag("")`, `p:option("foo")` without `-`, duplicate name, `p:argument("-bad")`) → raises via `error()`. These are programmer bugs.
- **`p:parse(...)`** never raises on user input. All user input problems are reported as `(nil, "message")` :
 - "unknown option '--xxx'"
 - "option '--xxx' requires a value"
 - "flag '-v' does not take a value"
 - "missing required option '--xxx'"
 - "missing required argument 'name'"
 - "argument 'name': invalid choice 'xxx'"
 - "option '--xxx': <converter message>"
 - "unexpected argument 'xxx'"
- A convert function that itself raises is **caught** by `parse()` and turned into a `(nil, "<...>: conversion error")`. The script never sees an exception coming out of `parse()`.

Design decisions

- **Help is a success, not a side effect.** Returning `{ help = true, usage = "..."}` instead of printing and exiting lets the script decide : print to stdout, log to a file, send to a GUI, prepend a banner, whatever. The library doesn't decide for you, consistent with the rest of LuaPilot where no module unilaterally exits the process.
- **(res, err) everywhere on user input, error() only on builder bugs.** Same convention as the rest of LuaPilot (user, json, sqlite, etc.).
- **Spec is a single string** ("`-v --verbose`"), not a chain of calls. Discoverable, terse, and avoids ambiguity when multiple names are declared together.
- **choices is checked on the raw string, then convert applied.** If you pair them, express choices as strings — e.g. `{ choices = {"1","2","3"}, convert = tonumber }`. Documented trap, but conscious : doing the reverse would force choices values to be of the converted type, which couples two independent features.
- **Positional with default is implicitly optional.** Saves the `required = false` boilerplate in the common case.
- **`parse()` reads `arg[1..n]`** — never `arg[0]` or negative indices, where LuaPilot puts its own slots. Just what the user actually typed.

Not in v1

Additive — could be added later without breaking SemVer :

- Sub-commands (`git commit / git push` style).
- Combined short options (`-abc` for `-a -b -c`).

- Short options with attached values (`-fvalue`).
- Variadic positionals (`nargs = "+"` or `"*"`).
- Argument groups (visual grouping in help).

[English](#) | [Français](#)

luapilot.exec — run an external program

Spawn a subprocess, capture its stdout / stderr, and get its exit code back as structured data. Where `os.execute` gives you just an integer, `luapilot.exec` gives you a result table with everything you actually need.

Why

Almost every non-trivial script eventually shells out — to `git`, `useradd`, `convert`, `ffmpeg`, whatever. `os.execute` is too crude (no captured output, shell quoting hazards), and `io.popen` doesn't let you both write to stdin and read stdout cleanly, doesn't expose the exit code without parsing, and has no timeout.

`luapilot.exec` is the structured alternative : argv passed as a list (no shell, no quoting bugs), output captured into memory with a bounded cap, optional timeout, separate stdout/stderr.

API

```
result, err = luapilot.exec(cmd, args?, opts?)
```

Argument	Type	Notes
<code>cmd</code>	string	Program to run. PATH is searched.
<code>args</code>	table (optional)	Array of string arguments — each becomes a separate argv element, no shell parsing.
<code>opts</code>	table (optional)	See below.

`opts` fields :

Field	Type	Default	Notes
<code>cwd</code>	string	inherit	Working directory for the child.
<code>env</code>	table {KEY = value, ...}	inherit	Merged with the parent environment, not replacing it. To unset a variable, omit it from <code>env</code> .
<code>stdin</code>	string	""	Data piped to the child's stdin.
<code>timeout</code>	number (s)	none	Kill the child with SIGKILL if it takes longer.
<code>max_output</code>	integer (bytes)	16 MiB	Hard cap per stream on captured output. Excess is silently dropped and signalled via <code>*_truncated</code> .

Result table on successful launch :

```
{
  stdout = "...",      -- captured stdout, up to max_output bytes
  stderr = "...",      -- captured stderr, up to max_output bytes
}
```

```

code = 0,                -- exit code (0..255)
timed_out = false,       -- true if killed by the timeout
stdout_truncated = false, -- true if more than max_output was produced
stderr_truncated = false,
}

```

Quick examples

```

-- Basic capture
local r = assert(luapilot.exec("git", { "rev-parse", "HEAD" }))
if r.code == 0 then
    print("HEAD:", r.stdout:gsub("\n$", ""))
end

-- Non-zero exit is NOT an error
local r = assert(luapilot.exec("grep", { "needle", "/etc/passwd" }))
if r.code == 0 then
    print("found:", r.stdout)
elseif r.code == 1 then
    print("not found")
else
    print("grep failed:", r.stderr)
end

-- Pipe data to stdin
local r = assert(luapilot.exec("sha256sum", { "-" }, {
    stdin = "hello world\n",
}))
print(r.stdout)
-- a948904f2f0f479b8f8197694b30184b0d2ed1c1cd2a1ec0fb85d299a192a447 -

-- Augment env (not replace)
local r = assert(luapilot.exec("env", nil, {
    env = { MY_FLAG = "yes" },
}))
-- Child sees MY_FLAG=yes plus everything the parent already had.

-- Timeout + truncation
local r = assert(luapilot.exec("yes", nil, {
    timeout = 0.5,
    max_output = 1024,
}))
print(r.timed_out, r.stdout_truncated) -- true true

-- Different cwd
local r = assert(luapilot.exec("ls", nil, { cwd = "/var/log" }))

```

Error contract

- **(nil, err)** only if the launch itself fails : program not found, cwd doesn't exist, fork failed, OOM, etc.
- **(result, nil)** for every other case, including :
 - The program ran and exited non-zero → `result.code > 0`.
 - The program was killed by the timeout → `result.timed_out` is true, `result.code` reflects the signal exit.
 - The program produced more output than `max_output` → the relevant `*_truncated` flag is true, captured data stops at the cap.
- **opts validation** : invalid types (`opts.cwd` not a string, `opts.env` not a table, `opts.env` keys with = or NUL, `opts.timeout` 0 or NaN/Inf, `opts.max_output` non-integer or > 2 GiB) → raise via `luaL_error`. These are

caller bugs.

Design decisions

- **Exit code error.** The most common bug with `os.execute` / `exec`-wrappers is treating non-zero exit as an exception. Many programs use exit codes for meaningful state (`grep` returns 1 when no match, `diff` returns 1 when files differ, etc.). The caller decides what counts as failure.
- **No shell.** `args` is an array, each element is one `argv` slot, no quoting needed and no `$(rm -rf $HOME)` surprises. If you really want a shell, pass `"sh" + { "-c", "your command" }`.
- **env merges with parent.** Replacing the full environment is rarely what you want and usually breaks programs that expect `PATH` or `LANG`. To override one variable, pass it. To unset one, pass it as the empty string and the child will see it empty (POSIX doesn't have "unset" semantics at `exec` time without extra effort).
- **max_output is per stream, not total.** 16 MiB stdout + 16 MiB stderr is the cap. Large enough to capture most build logs, small enough that a runaway subprocess can't OOM the `LuaPilot` process.
- **Hard 2 GiB cap on max_output.** Pretty much anything that needs more than 2 GiB of captured stdout should redirect to a file via `opts.stdin` + a shell wrapper, or via `exec("sh", { "-c", "your-cmd > /tmp/log" })`.
- **stdin is a string, not a callback.** Streaming gigabytes into a child is rare and out of scope ; the common case is "feed this small input and read the answer".

Not in v1

- Streaming stdout / stderr (line-by-line callback during execution). Currently the whole capture is returned at once.
- Stdin streaming (callback that produces input chunks).
- Direct fd redirection (`stdout = "/tmp/log"`). Easy to add later if needed ; the shell-wrapper workaround above covers most cases.
- Process group management beyond the child. The current implementation does correctly tear down the child's process group on timeout / interrupt — see [security](#).

[English](#) | [Français](#)

luapilot fs — file system

A flat set of file system operations exposed directly on `luapilot` (no sub-namespace, predates the convention). Covers the everyday needs that Lua's `os.*` and `io.*` don't : recursive listing, copy with attributes, hashing, path manipulation, attributes, links.

Why

Lua's standard `io` and `os` give you `open`, `rename`, `remove`, and that's about it. Everything beyond that — listing a directory, recursive copy, computing a hash, querying `mtime` — requires external programs or low-level FFI. `luapilot` collects these in a single set of functions that behave consistently (paths are strings, errors come back as `(nil, "...")`).

All paths are accepted as plain strings. Tilde expansion (`~/x`) is **not** automatic — the caller must resolve `$HOME` itself if needed. Functions never expand globs ; for glob support, use `luapilot.find` with an explicit pattern.

API — Existence, type, attributes

Function	Returns
<code>luapilot.fileExists(path)</code>	<code>boolean</code>
<code>luapilot.isFile(path)</code>	<code>boolean</code> (true only for regular files)
<code>luapilot.isDir(path)</code>	<code>boolean</code>
<code>luapilot.fileSize(path)</code>	<code>integer bytes</code> <code>(nil, err)</code>

Function	Returns
<code>luapilot.getAttributes(path)</code>	table with <code>mtime</code> , <code>mode</code> , <code>size</code> , <code>uid</code> , <code>gid</code> , <code>inode</code> , etc. <code>(nil, err)</code>
<code>luapilot.setAttributes(path, uid, gid, mode?)</code>	<code>(true, nil)</code> <code>(nil, err)</code> — set owner/group, optionally mode
<code>luapilot.symlinkattr(path)</code>	same shape as <code>getAttributes</code> , but <code>lstat</code> (does not follow symlinks) <code>(nil, err)</code>
<code>luapilot.getMode(path)</code>	octal integer <code>(nil, err)</code>
<code>luapilot.setMode(path, mode)</code>	<code>(true, nil)</code> <code>(nil, err)</code> — mode is an integer (use <code>0xled</code> for <code>0755</code> , or build it with <code>tonumber("755", 8)</code>)

API — Creating, removing, moving

Function	Returns
<code>luapilot.touch(path)</code>	<code>(true, nil)</code> <code>(nil, err)</code> — create file or update <code>mtime</code>
<code>luapilot.mkdir(path, opts?)</code>	<code>(true, nil)</code> <code>(nil, err)</code> — <code>opts.parents = true</code> for <code>mkdir -p</code>
<code>luapilot.rmdir(path)</code>	<code>(true, nil)</code> <code>(nil, err)</code> — only empty dirs
<code>luapilot.rmdirAll(path)</code>	<code>(true, nil)</code> <code>(nil, err)</code> — recursive
<code>luapilot.remove(path)</code>	<code>(true, nil)</code> <code>(nil, err)</code> — file or symlink
<code>luapilot.rename(old, new)</code>	<code>(true, nil)</code> <code>(nil, err)</code>
<code>luapilot.chdir(path)</code>	<code>(true, nil)</code> <code>(nil, err)</code>
<code>luapilot.currentDir()</code>	string (absolute)
<code>luapilot.joinPath(a, b, ...)</code>	string — like <code>path/a/b/c</code> ; also accepts a single table of segments
<code>luapilot.link(target, link, opts?)</code>	<code>(true, nil)</code> <code>(nil, err)</code> — <code>opts.symbolic = true</code> for symlinks

API — Path manipulation

These operate on path *strings* — they don't touch the filesystem.

Function	Returns	Example
<code>luapilot.getBasename(path)</code>	string <code>(nil, err)</code> — last component	<code>"/etc/hostname" → "hostname"</code>
<code>luapilot.getPath(path)</code>	string <code>(nil, err)</code> — parent dir (dirname)	<code>"/etc/hostname" → "/etc"</code>
<code>luapilot.getFilename(path)</code>	string <code>(nil, err)</code> — basename without extension	<code>"report.tar.gz" → "report.tar"</code>
<code>luapilot.getExtension(path)</code>	string <code>(nil, err)</code> — final extension	<code>"report.tar.gz" → ".gz"</code>

API — Listing and finding

Function	Returns
<code>luapilot.listFiles(path)</code>	table (regular files in path) <code>(nil, err)</code>
<code>luapilot.find(path, opts)</code>	iterator function, see below
<code>luapilot.createFileIterator(path)</code>	iterator function over directory entries

`find` accepts `opts` :

- `pattern = "*.lua"` — shell glob applied to file name only.
- `recursive = true` — descend into sub-directories.
- `type = "f" / "d"` — restrict to files or dirs.
- `max_depth = N` — limit recursion depth.

API — Copy

Function	Returns
<code>luapilot.copy(src, dst, opts?)</code>	<code>(true, nil) (nil, err)</code>
<code>luapilot.copyTree(src, dst, opts?)</code>	<code>(true, nil) (nil, err)</code> — recursive
<code>luapilot.moveTree(src, dst)</code>	<code>(true, nil) (nil, err)</code>

copy options :

- `preserve = true` — keep mtime, mode, owner where possible.
- `overwrite = true` — replace destination if it exists.

API — Hashes and checksums

Hash digests of file contents. Result is **lowercase hex string** unless noted. Note the `sum` suffix — these are the registered names in `luapilot.*`.

Function	Algorithm	Notes
<code>luapilot.crc32sum(path)</code>	CRC-32 (IEEE 802.3)	Not cryptographic. For integrity / accidental-error detection only.
<code>luapilot.md5sum(path)</code>	MD5	Legacy. Don't use for security.
<code>luapilot.sha1sum(path)</code>	SHA-1	Legacy. Don't use for security.
<code>luapilot.sha256sum(path)</code>	SHA-256	
<code>luapilot.sha384sum(path)</code>	SHA-384	
<code>luapilot.sha512sum(path)</code>	SHA-512	
<code>luapilot.sha3_256sum(path)</code>	SHA3-256	
<code>luapilot.sha3_384sum(path)</code>	SHA3-384	
<code>luapilot.sha3_512sum(path)</code>	SHA3-512	
<code>luapilot.blake2b512sum(path)</code>	BLAKE2b-512	

Each returns `string (hex)` or `(nil, err)`. The file is streamed, not loaded entirely into RAM. Non-regular files (directories, sockets, ...) are rejected with `(nil, "<algo>: not a regular file")`.

MD5 and SHA-1 are exposed for compatibility (Git, legacy systems) — don't use them for new cryptographic uses. **CRC-32 is even further from cryptographic**: an attacker can produce collisions trivially. Use it for integrity / change detection, never for authentication.

API — In-memory checksums

Compute a hash of bytes already in memory, without round-tripping through a temporary file. Binary-safe (embedded NULs are allowed).

Function	Returns
<code>luapilot.crc32(data)</code>	<code>string</code> — 8-char lowercase hex

Cannot fail at runtime (pure computation). Wrong argument types raise via `luaL_error`. As with `crc32sum`, **not cryptographic**.

Quick examples

```
-- Predicates
if luapilot.isdir("/etc") and luapilot.fileExists("/etc/hostname") then
    print("size:", luapilot.fileSize("/etc/hostname"))
end

-- Path manipulation (pure string ops)
local base = luapilot.getBaseName("/var/log/syslog.1") -- "syslog.1"
local dir  = luapilot.getPath("/var/log/syslog.1")     -- "/var/log"
local ext  = luapilot.getExtension("/var/log/syslog.1") -- ".1"

-- Recursive listing
for entry in luapilot.find("/var/log",
    { pattern = "*.log", type = "f", recursive = true }) do
    print(entry)
end

-- Copy with attributes
luapilot.copyTree("./src", "/tmp/backup",
    { preserve = true, overwrite = true })

-- Hash a release tarball
local sha, err = luapilot.sha256sum("/tmp/luapilot-1.7.0.tar.gz")
if sha then print("SHA256:", sha) end

-- Quick integrity check (NOT cryptographic, but fast)
local crc = luapilot.crc32sum("/tmp/luapilot-1.7.0.tar.gz")
print("CRC32:", crc) -- e.g. "a1b2c3d4"

-- In-memory CRC32 of arbitrary bytes (binary-safe, NULs OK)
print(luapilot.crc32("abc")) -- "352441c2"
print(luapilot.crc32("")) -- "00000000"

-- Set ownership and permissions
local mode_755 = tonumber("755", 8) -- octal -> integer
assert(luapilot.setMode("/srv/myapp/run.sh", mode_755))
-- root only:
-- assert(luapilot.setAttributes("/srv/myapp", 1000, 1000))
```

Error contract

- All functions follow the `(nil, err)` convention on runtime failure (no such file, permission denied, etc.).
- Wrong argument **types** raise via `luaL_error`.
- **setAttributes(path, uid, gid)** rejects negative UIDs/GIDs and values above `uid_t/gid_t` max (typically $2^{32} - 1$). Without that upper-bound check, a value above the limit would truncate silently — worst case to UID 0 (root). Same logic as the `user` module.
- Path safety : `copy`, `copyTree`, `moveTree` use `lexically_relative()` and component-wise `is_within()` guards to prevent symlink-based escapes out of the destination tree. See [security](#).

Design decisions

- **Flat namespace** : `fileExists` instead of `fs.exists`, `getAttributes` instead of `fs.attr.get`. Pure historical reason — these functions predate the per-module convention. Stable now ; renaming would break every script.
- **get* / set* for path attributes** (`getMode/setMode`, `getAttributes/setAttributes`) but plain verbs for FS actions (`remove`, `touch`, `mkdir`). The “get/set” pairs exist because both directions are needed ; the others are single-direction operations.

- **Path manipulation is separate from FS access.** `getBasename`, `getPath`, etc. operate on strings only — they don't stat the path. Use `fileExists` if you need to know whether the path actually exists.
- **Hashes have a `sum` suffix** to match the standard Unix tools (`md5sum`, `sha256sum`, etc.). Discoverable for anyone familiar with the command line.
- **`copy` is single-file, `copyTree` is recursive.** Lua doesn't have function overloading by type, so splitting them avoids the ambiguity “did `copy('dir', 'dir')` mean recursive or not?”.

Not in v1

- Glob matching as a standalone function (`luapilot.glob`). Use `find` with `pattern = "*.lua"` for the common case.
- Async I/O. Use [workers](#) for parallel file processing instead.
- BLAKE2s (only BLAKE2b-512 is exposed via OpenSSL EVP).

[English](#) | [Français](#)

luapilot.http — HTTP client

A simple HTTP/HTTPS client built on [cpp-httplib](#), reusing the vendored OpenSSL. Synchronous, blocking, with timeouts.

Why

Almost every non-trivial script needs to talk to some HTTP API. Without a built-in client, you reach for `curl` via `exec`, with all the quoting and process-spawning overhead. `luapilot.http` makes a request a one-liner, with proper TLS verification by default.

API

Function	Returns
<code>luapilot.http.request(opts)</code>	<code>response (table) (nil, err)</code>
<code>luapilot.http.get(url, opts?)</code>	shortcut for <code>request{ method="GET", url=url, ... }</code>
<code>luapilot.http.post(url, opts?)</code>	shortcut for <code>request{ method="POST", url=url, ... }</code>
<code>luapilot.http.put(url, opts?)</code>	shortcut for <code>request{ method="PUT", url=url, ... }</code>
<code>luapilot.http.delete(url, opts?)</code>	shortcut for <code>request{ method="DELETE", url=url, ... }</code>

opts table

Field	Type	Default
<code>url</code>	string (required for request)	—
<code>method</code>	string	"GET"
<code>headers</code>	table of <code>name = value</code>	{}
<code>body</code>	string	""
<code>timeout</code>	number (seconds)	30
<code>verify</code>	boolean (TLS cert verification)	true
<code>ca_cert</code>	string (path to CA bundle file)	system default
<code>ca_path</code>	string (path to CA dir)	system default
<code>follow_redirects</code>	boolean	true
<code>max_redirects</code>	integer	5

response table

```
{
  status = 200,
```

```

body = "...",
headers = {
    -- normalised lowercase keys
    ["content-type"] = "application/json",
    ["content-length"] = "42",
},
-- final URL after redirects:
url = "https://api.example.com/v1/data",
}

```

Quick examples

```

-- Simple GET
local r, err = luapilot.http.get("https://api.example.com/health")
if r then
    print(r.status, r.body)
end

-- JSON POST with auth header
local r, err = luapilot.http.post("https://api.example.com/v1/things", {
    headers = {
        ["Content-Type"] = "application/json",
        ["Authorization"] = "Bearer " .. token,
    },
    body = luapilot.json.encode({ name = "widget", count = 7 }),
    timeout = 10,
})

-- Self-signed cert (dev / private CA)
local r = luapilot.http.get("https://internal.svc/", {
    ca_cert = "/etc/myapp/internal-ca.crt",
})

-- Disable verification (TESTING ONLY)
local r = luapilot.http.get("https://expired.badssl.com/",
    { verify = false })

```

Error contract

- **Wrong argument types** → raises via `luaL_error`.
- **Network errors** (DNS, connect, timeout, TLS) → `(nil, "http: <description>")`.
- **HTTP status errors** are **not** errors — a 404 returns a normal response table with `status = 404`. Status semantics are the caller's responsibility.

TLS / trust store

LuaPilot ships its own static OpenSSL, so it doesn't automatically inherit the distro's `ca-certificates` configuration. Two mechanisms ensure `verify=true` works out of the box :

1. The vendored OpenSSL is built with `--openssldir=/etc/ssl`, covering Arch, Debian, Ubuntu, Alpine, Gentoo.
2. At TLS init, LuaPilot probes known CA bundle paths for Fedora/RHEL, OpenSUSE, FreeBSD, NetBSD.

You can override per-call with `ca_cert` / `ca_path`, or globally via `SSL_CERT_FILE` / `SSL_CERT_DIR` environment variables. See [security](#) and [tls](#) for details.

Design decisions

- **Synchronous, blocking.** Scripts are usually one-shot request-response affairs ; sync is simpler and sufficient. For many parallel calls, use [workers](#).

- **verify=true by default.** Disabling TLS verification must be explicit (`verify=false`). No silent downgrade.
- **HTTP status is not an error.** The caller decides whether 404 or 500 is failure ; the network call itself succeeded.
- **Headers are normalised to lowercase keys** in the response, to make case-insensitive lookups work directly (`r.headers["content-type"]`).
- **Follow redirects by default.** Bounded to 5 hops, override with `max_redirects` or disable with `follow_redirects=false`.

Not in v1

- Streaming response body (e.g. for downloading large files). Currently `body` is read fully into memory.
- HTTP/2 / HTTP/3.
- WebSockets (use `socket` + TLS + a Lua framing library if needed).
- Server side. `cpp-httplib` supports it, but exposing a robust HTTP server to Lua is its own design effort.

[English](#) | [Français](#)

luapilot.inotify — filesystem event watching

Wraps Linux `inotify(7)` for instant filesystem event delivery — no polling. Used for hot-reloading configuration, watching drop directories, log tailing, file synchronisation triggers, etc.

Why

Polling a directory with `listDir` every second is wasteful (CPU + inode lookups) and laggy. `inotify` is the kernel's purpose-built mechanism : the kernel wakes your script only when something actually changes, with sub-millisecond latency.

The Lua API exposes a minimal contract over a notoriously tricky syscall. Queue overflow is surfaced explicitly (the only thing worse than dropping events silently is not telling you about it).

API

Function	Returns
<code>luapilot.inotify.new()</code>	<code>watcher (userdata) (nil, err)</code>
<code>w:add(path, events, opts?)</code>	<code>integer wd (watch descriptor) (nil, err)</code>
<code>w:remove(wd)</code>	<code>(true, nil) (nil, err)</code>
<code>w:read(timeout?)</code>	<code>table of events (nil, "timeout") (nil, "interrupted") (nil, err)</code>
<code>w:close()</code>	<code>(true, nil) — idempotent</code>

events argument

A strict 1..N array of event name strings. Acceptable names :

Name	Triggered when
"access"	file accessed
"modify"	file content modified
"attrib"	metadata changed (perms, mtime, ...)
"close_write"	file opened for writing is closed
"close_nowrite"	read-only fd closed
"open"	file opened
"moved_from"	file moved out of watched dir
"moved_to"	file moved into watched dir
"create"	file/dir created in watched dir

Name	Triggered when
"delete"	file/dir deleted from watched dir
"delete_self"	the watched path itself is deleted
"move_self"	the watched path is moved

Event returned by read

```
{
  wd      = 1,                -- watch descriptor
  name    = "newfile.txt",    -- "" if the watched path itself is the target
  events  = { create = true, close_write = true },
  is_dir  = false,
  cookie  = 0,                -- non-zero pairs moved_from / moved_to
}
```

Special synthetic event for **queue overflow** :

```
{ wd = -1, events = { overflow = true } }
```

When you see this, the kernel queue ran out of space and some events were lost. Re-scan the watched paths to recover state.

Quick example

```
local w = assert(luapilot.inotify.new())
assert(w:add("/srv/incoming", { "close_write", "moved_to" }))

luapilot.signal.handle("TERM", function() w:close(); os.exit(0) end)

while true do
  local events, err = w:read()    -- blocks until something happens
  if not events then
    if err == "interrupted" then break end
    io.stderr:write("inotify: ", err, "\n"); break
  end
  for _, ev in ipairs(events) do
    if ev.events.overflow then
      rescan()                    -- queue overflowed, recover
    else
      handle_new_file("/srv/incoming/" .. ev.name)
    end
  end
end
```

Error contract

- **read(t)** where t is NaN/Inf/negative → raises via `luaL_error`.
- **add(path, events)** with a non-list `events` table (sparse, extra keys, non-string elements) → (nil, err).
- **add on a non-existent path** → (nil, "no such file or directory").
- **read** :
 - events table on success.
 - (nil, "timeout") if timeout elapsed.
 - (nil, "interrupted") if a handled signal arrived.
 - (nil, err) on other errors.
- **Methods after close** → (nil, "inotify: closed").

Design decisions

- **Non-recursive in v1.** `inotify(7)` itself isn't recursive ; watching a tree means walking it and calling `add` on each sub-directory, then handling `create` events to extend the watch. That's a real design effort with subtleties (races between walking and event delivery, overflow handling) and worth a dedicated module if/when needed. Buildable in pure Lua on top.
- **Events list mandatory, no implicit “all”.** Watching everything floods the queue and forces every event to traverse Lua. Forcing the caller to pick events makes the cost visible.
- **Overflow surfaced explicitly**, never swallowed. The kernel has a finite queue (`/proc/sys/fs/inotify/max_queued_events`, default 16384). When it overflows, events are lost. The only appropriate action is to re-scan ; silently dropping the signal would defeat the entire reliability story.
- **`read()` is interruptible by signal.** A `read()` blocking forever ignoring `SIGTERM` is the classic graceful-shutdown bug. See [signal](#).
- **`IN_CLOEXEC` on the fd :** not inherited by `exec'd` subprocesses, consistent with sockets.

Not in v1

- Recursive watching. Add at the script level (`walk + add`) or wait for a `recursive = true` option.
- Multiple watchers per process. Doable today by creating multiple `inotify.new()` instances ; not a built-in registry.
- `IN_MASK_ADD` semantics (adding events to an existing watch). Currently `add(path, events)` replaces the mask. Rare need.

[English](#) | [Français](#)

luapilot.json — JSON encode/decode

Backed by [nlohmann/json](#), the most widely-used C++ JSON library. Handles the usual round-trip plus the explicit “empty array” sentinel that Lua tables can't disambiguate from “empty object”.

Why

The Lua/JSON impedance mismatch always trips someone up : Lua has no distinction between an empty list and an empty table. Most ad-hoc Lua JSON libraries pick one and get it wrong half the time. `luapilot.json` is explicit : `{}` decodes to `{}` and re-encodes to `{}` (object), but if you want `[]` you use the `luapilot.json.empty_array` sentinel.

API

Function	Returns
<code>luapilot.json.encode(value, opts?)</code>	<code>string</code> (JSON text) <code>(nil, err)</code>
<code>luapilot.json.decode(text)</code>	<code>value</code> <code>(nil, err)</code>
<code>luapilot.json.empty_array</code>	sentinel value that encodes to <code>[]</code>

Encoding options (opts table) :

- `pretty = true` — pretty-print with indentation (default `false`).
- `indent = N` — indent width when `pretty=true` (default 2).

Quick example

```
local J = luapilot.json

-- Decode
local data, err = J.decode([[
{ "name": "alice", "tags": ["admin", "ops"], "active": true }
]])
```

```

print(data.name, data.tags[1])

-- Encode (compact)
local out = J.encode({ a = 1, b = { 2, 3 } })
-- out == '{"a":1,"b":[2,3]}'

-- Encode pretty
local pretty = J.encode({ a = 1 }, { pretty = true, indent = 4 })

-- Empty array vs empty object
J.encode({}) --> "{}"
J.encode(J.empty_array) --> "[]"
J.encode({ items = J.empty_array }) --> '{"items":[]}'

```

Error contract

- **encode** : (nil, err) on encoding errors (cycle in the graph, value that doesn't map to JSON like a function or userdata, NaN/Inf numbers, etc.).
- **decode** : (nil, err) on parse error. The error message includes the line/column where parsing failed.
- **Wrong argument types** → raises via `luaL_error`.

Design decisions

- **empty_array sentinel** instead of guessing from the table structure (some libraries treat `{1, 2, 3}` as array and `{a=1}` as object, which is correct, but `{}` is ambiguous). An explicit sentinel removes all surprise.
- **NaN/Inf rejected**, not silently encoded as `null`. Different consumers handle JSON `null` differently for numeric fields ; better to fail loudly and let the caller decide.
- **No streaming API**. Round-trips are bounded by your script's memory regardless ; if you need to handle JSON larger than RAM, reach for a real streaming parser in a separate process.

Not in v1

- Streaming encoder/decoder.
- A schema validator (use a pure-Lua one — they're trivial to add at the script level).

English | [Français](#)

logging — leveled logger (Lua module)

A standard leveled logger : `debug`, `info`, `warn`, `error`, `fatal`. Bundled as a pure-Lua module, loaded via `require("logging")`. Not in the `luapilot` namespace — it's a library module, like `inspect`.

Why

Every non-trivial script ends up reinventing some flavour of “levels + timestamps + optional file output”. Standardising it removes the bikeshed and makes log output consistent across LuaPilot scripts.

API

```
local log = require("logging")
```

Function	Returns
<code>log.set_level(name)</code>	(true, nil) — name is one of "debug", "info", "warn", "error", "fatal"
<code>log.set_output(path)</code>	(true, nil) (nil, err) — file path (default stderr)

`log.debug(...), log.info(...), log.warn(...),` each prints if its level is `>` current threshold
`log.error(...), log.fatal(...)`

Messages are timestamped (YYYY-MM-DD HH:MM:SS) and tagged with the level ([DEBUG], [INFO], etc.). Multiple arguments are concatenated with spaces, like `print`.

Quick example

```
local log = require("logging")
log.set_level("info")  -- debug calls below this become no-ops

log.info("startup, pid =", luapilot.pid())
log.warn("retry attempt", 3, "of", 5)
log.error("connection failed:", err)

-- Optional: route to a file
local ok, err = log.set_output("/var/log/myapp.log")
if not ok then
    log.fatal("can't open log:", err)
    os.exit(1)
end
```

Output :

```
2026-06-11 14:32:01 [INFO] startup, pid = 12345
2026-06-11 14:32:05 [WARN] retry attempt 3 of 5
2026-06-11 14:32:05 [ERROR] connection failed: timeout
```

Error contract

- **set_level(name)** : unknown name raises via `error()`.
- **set_output(path)** : (nil, err) if the file can't be opened for append. Logging continues to the previous sink (stderr if none was set).
- Logging functions themselves never fail — if the file becomes unwritable mid-run, the message is silently dropped (the alternative — crashing the script because the log volume is full — is worse).

Design decisions

- **Pure-Lua module.** No C++ needed, and being in Lua means scripts can monkey-patch it (e.g. add JSON-format output) at the call site without rebuilding LuaPilot.
- **Five levels, no trace.** Five is enough for almost everyone ; adding more invites everyone to invent their own.
- **Single global sink.** Multiple loggers / hierarchical loggers / per-module levels are a feature creep trap. If you need multiple destinations, write a wrapper.

Not in v1

- JSON / structured logging output. Easy to bolt on at the script level if needed.
- Log rotation. Use `logrotate(8)` on the file output.
- Async / buffered output. Premature optimisation for typical use.

English | [Français](#)

luapilot.signal — POSIX signals

Register Lua callbacks for POSIX signals (SIGTERM, SIGINT, SIGHUP, ...) so long-running scripts can shut down gracefully or reload configuration on demand.

Why

Long-running LuaPilot scripts (bots, daemons, watchers) need to handle signals properly :

- **SIGTERM** from **systemctl stop** should trigger a clean shutdown.
- **SIGINT** from Ctrl-C should do the same.
- **SIGHUP** is the conventional “reload config” signal.

Without explicit handlers, these signals just kill the process, leaving open files, half-written databases, and orphaned children.

API

Function	Returns
<code>luapilot.signal.handle(name, fn)</code>	<code>(true, nil) (nil, err)</code> — install Lua callback
<code>luapilot.signal.ignore(name)</code>	<code>(true, nil) (nil, err)</code> — set to SIG_IGN
<code>luapilot.signal.default(name)</code>	<code>(true, nil) (nil, err)</code> — back to OS default
<code>luapilot.signal.kill(pid, name)</code>	<code>(true, nil) (nil, err)</code> — send signal to PID
<code>luapilot.signal.list()</code>	table of all supported signal names
<code>luapilot.signal.is_pending()</code>	boolean — any handled signal queued ?

Signal names accepted (string, case-sensitive) :

"HUP", "INT", "QUIT", "USR1", "USR2", "PIPE", "ALRM", "TERM", "CHLD", "CONT", "TSTP", "TTIN", "TTOU", "WINCH". Other signals (KILL, STOP) cannot be caught — POSIX forbids it.

Quick example

```
local sig = luapilot.signal
local running = true

sig.handle("TERM", function()
    print("got SIGTERM, shutting down")
    running = false
end)
sig.handle("INT", function()
    print("got SIGINT, shutting down")
    running = false
end)
sig.handle("HUP", function()
    print("got SIGHUP, reloading config")
    cfg = load_config()
end)

while running do
    local line, err = sock:recv_line(1.0)
    if not line then
        if err == "interrupted" then
            -- A handled signal arrived during recv ; the callback
            -- already fired. Re-check the loop condition.
        else
            break
        end
    else
        process(line)
    end
end
```

```
print("clean exit")
```

How callbacks run

When a signal arrives :

1. The kernel sets a pending flag (no Lua code runs from the signal handler itself — async-signal-safe restrictions).
2. If the script is in a blocking call (`recv`, `sleep`, `inotify.read`, `accept`, ...), the call returns `(nil, "interrupted")`.
3. Before returning, LuaPilot dispatches all pending callbacks in registration order, in the main Lua thread.
4. The script continues normally — the callback can have modified globals (`running = false` is the typical pattern).

This means callbacks always run in Lua context, never from inside a signal handler. They can use any Lua features (no async-signal- safety restrictions).

Error contract

- **Unknown signal name** → `(nil, "signal: unknown name 'XYZ'")`.
- **Calls forbidden inside a worker thread** → `(nil, "signal: not available in worker threads")`. Signals are process-global ; only the main thread manages them.
- **Wrong argument types** → raises via `luaL_error`.
- **kill(pid, name)** for a PID that doesn't exist or isn't yours → `(nil, "kill: ...")`.

Design decisions

- **Callbacks run in the Lua main thread, not from the signal handler.** POSIX async-signal-safety rules forbid most useful operations from inside a real handler. By marking the signal as pending and dispatching from the next safe point, callbacks can do anything Lua can do.
- **Blocking calls return "interrupted" on a handled signal.** Without this, a `recv_line` waiting on the network would block until timeout regardless of `SIGTERM` arriving. With it, the shutdown can complete in milliseconds.
- **No re-entrance.** While a callback is running, additional signals are queued and dispatched after.
- **Forbidden in worker threads.** Signals are process-wide ; only one thread can sensibly own them. Workers must use channels for inter-thread communication.

Not in v1

- `sigprocmask` / fine-grained signal masking. Not commonly needed at script level.
- `signalfd`-based polling integration. The current dispatch model is simpler and covers the typical cases.

[English](#) | [Français](#)

luapilot.socket — TCP client and server

Low-level TCP sockets with timeouts, signal awareness, and a small set of higher-level helpers (`recv_line`, `recv_all`). Foundation for any custom protocol — the IRC bot, a JSON-over-TCP daemon, etc. See [tls](#) for the encrypted variant.

Why

For everyday HTTP needs there's [http](#). But many scripts need to speak a custom line-oriented protocol (IRC, Redis, plain telnet-style daemons), where a real TCP socket with timeouts is the right tool. Lua has no built-in TCP, and `LuaSocket` is a separate dep ; `luapilot.socket` makes the common patterns one-liners.

API

Constructors

Function	Returns
<code>luapilot.socket.connect(host, port, opts?)</code>	<code>socket</code> <code>(nil, err)</code>
<code>luapilot.socket.listen(host, port, opts?)</code>	<code>server_socket</code> <code>(nil, err)</code>

`opts` :

- `timeout = N` — default timeout in seconds for blocking I/O (default infinite).
- `connect_timeout = N` — only for `connect`, default 30 s.
- `nodelay = true` — set `TCP_NODELAY`.
- `keepalive = true` — set `SO_KEEPALIVE`.

Methods (client socket)

Method	Returns
<code>s:send(data)</code>	integer bytes sent <code>(nil, err)</code>
<code>s:recv(n, timeout?)</code>	string <code>(nil, "timeout")</code> <code>(nil, "interrupted")</code> <code>(nil, err)</code>
<code>s:recv_line(timeout?)</code>	string (without <code>\n</code>) <code>(nil, ...)</code>
<code>s:recv_all(timeout?)</code>	string (read until EOF) <code>(nil, ...)</code>
<code>s:set_timeout(seconds)</code>	sets default timeout (<code>0</code> = infinite)
<code>s:close()</code>	idempotent
<code>s:peer()</code>	<code>(host, port)</code> <code>(nil, err)</code> — remote endpoint

`recv_line` reads bytes until `\n` (LF) is found. CR is stripped if present (`\r\n` → returned without the trailing `\r`). Has a hard 8 MiB cap to prevent DoS via an endless single line.

Methods (server socket)

Method	Returns
<code>srv:accept(timeout?)</code>	<code>socket</code> <code>(nil, ...)</code>
<code>srv:close()</code>	idempotent

Quick examples

TCP echo client

```
local s = assert(luapilot.socket.connect("localhost", 4000, {
    timeout = 5,
    nodelay = true,
}))

s:send("hello\n")
local reply, err = s:recv_line()
print("got:", reply)
s:close()
```

Simple line-protocol server with graceful shutdown

```
local sig = luapilot.signal
local srv = assert(luapilot.socket.listen("0.0.0.0", 4000))
local running = true
sig.handle("TERM", function() running = false end)
sig.handle("INT", function() running = false end)
```

```

while running do
    local client, err = srv:accept(1.0)      -- 1 s timeout
    if not client then
        if err == "timeout" or err == "interrupted" then
            -- loop and re-check `running`
        else
            io.stderr:write("accept: ", err, "\n"); break
        end
    else
        repeat
            local line = client:recv_line(30)
            if line then client:send("you said: " .. line .. "\n") end
        until not line
        client:close()
    end
end
srv:close()

```

Error contract

- **Connect errors** (DNS, refused, timeout) → (nil, "socket: ...").
- **Bind errors** (port in use, permission) → (nil, "socket: ...").
- **recv** :
 - string (any length up to n requested).
 - Empty string "" when the peer closed cleanly. Not an error.
 - (nil, "timeout") if no data within timeout.
 - (nil, "interrupted") if a handled signal arrived.
 - (nil, "line too long") for recv_line past the 8 MiB cap.
 - (nil, err) on other errors.
- **send** : may return fewer bytes than requested. Wrap in a loop if you need to send a fixed amount (or use recv_all- symmetric semantics in your protocol).
- **Methods after close** → (nil, "socket: closed").
- **NaN/Inf timeouts**, negative timeouts → raises via luaL_error.

Design decisions

- **Single-threaded, blocking I/O.** No select/poll exposed — use [workers](#) for concurrent connections, or short timeouts in a polling loop for simple cases.
- **recv_line has an 8 MiB cap.** Protects against a peer that sends megabytes without ever sending \n. Configurable later if a use case needs more.
- **Signal-aware blocking.** All recv* and accept return "interrupted" on a handled signal, so the script can exit cleanly.
- **recv returns "" on clean close, not nil.** Lets scripts distinguish “remote went away” (“”) from “no data yet” (“timeout”) without inventing yet another sentinel.

Not in v1

- UDP sockets. Easy to add additively (luapilot.socket.udp.*).
- Unix-domain sockets. Same reasoning.
- Native multiplexing (select/poll/epoll exposed to Lua). Use workers or short-timeout polling instead.

English | [Français](#)

luapilot.sqlite — embedded SQL database

Wraps SQLite 3.53.1 (vendored), the most-deployed database engine in the world. Zero-config persistence for any script that needs more than a JSON file but less than a database server.

Why

A script that needs to keep state across runs (a bot's seen-list, a scraper's progress, accumulated metrics) shouldn't have to pull in PostgreSQL or invent its own file format. SQLite is exactly right at this scale : single file, transactional, fast, no daemon.

The Lua API mirrors the SQLite C API closely (open / exec / prepare / step / finalize) with safer defaults and Lua-friendly error returns.

API

Opening and closing

Function	Returns
<code>luapilot.sqlite.open(path, opts?)</code>	<code>db (userdata) (nil, err)</code>
<code>db:close()</code>	<code>(true, nil)</code> — idempotent

`opts` :

Field	Type	Default
<code>readonly</code>	boolean	false
<code>wal</code>	boolean — enable WAL journal mode	false
<code>busy_timeout</code>	number ms (block this long on a locked db)	5000
<code>foreign_keys</code>	boolean	true

Special path `":memory:"` opens an in-memory database (lost on close). Use for tests or transient processing.

Direct exec (no parameters / one-shot statement)

Function	Returns
<code>db:exec(sql)</code>	<code>(true, nil) (nil, err)</code>
<code>db:exec(sql, params)</code>	same, with ?-bound parameters
<code>db:query(sql, params?)</code>	table of row tables <code>(nil, err)</code>

Prepared statements (re-use, performance)

Function	Returns
<code>db:prepare(sql)</code>	<code>stmt (userdata) (nil, err)</code>
<code>stmt(params?)</code>	iterator over rows
<code>stmt:exec(params?)</code>	<code>(true, nil) (nil, err)</code>
<code>stmt:finalize()</code>	<code>(true, nil)</code> — idempotent

Transactions

Function	Returns
<code>db:transaction(fn)</code>	<code>(true, result)</code> if <code>fn</code> returns truthy ; rollback + <code>(false, err)</code> otherwise

Utilities

Function	Returns
<code>db:last_insert_rowid()</code>	integer
<code>db:changes()</code>	integer — rows affected by last write
<code>db:in_transaction()</code>	boolean

Type mapping

SQLite type	Lua type
INTEGER	integer
REAL	number (float)
TEXT	string
BLOB	string (binary-safe)
NULL	nil (read), nil or <code>db.NULL</code> sentinel (write)

Quick examples

```

local db = assert(luapilot.sqlite.open("state.db", { wal = true }))

-- Schema
db:exec([[
    CREATE TABLE IF NOT EXISTS users (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        name TEXT UNIQUE NOT NULL,
        active INTEGER DEFAULT 1
    );
]])

-- Insert with parameters
assert(db:exec("INSERT INTO users (name) VALUES (?)", { "alice" }))
print("inserted as id", db:last_insert_rowid())

-- Query rows
local rows = assert(db:query("SELECT id, name FROM users WHERE active = ?", { 1 }))
for _, r in ipairs(rows) do
    print(r.id, r.name)
end

-- Prepared statement reused in a loop (fast path)
local stmt = assert(db:prepare("INSERT INTO users (name) VALUES (?)"))
for _, name in ipairs({ "bob", "carol", "dave" }) do
    assert(stmt:exec({ name }))
end
stmt:finalize()

-- Transaction
local ok, err = db:transaction(function()

```

```

db:exec("UPDATE users SET active = 0 WHERE name = ?", { "alice" })
db:exec("UPDATE users SET active = 1 WHERE name = ?", { "bob" })
return true    -- commit
end)
if not ok then print("rollback:", err) end

db:close()

```

Error contract

- All runtime errors → (nil, "sqlite: <description>") with the SQLite error code in the message.
- **exec(sql, params)** with ? placeholders but no params argument → (nil, "sqlite: placeholders found but no params table provided") — explicit error rather than silent NULL binding, which is a classic injection footgun.
- **Sparse keys in params** (e.g. {[1] = "a", [3] = "c"}) → (nil, err).
- **Empty BLOB string** → stored correctly as empty BLOB (previously buggy in pre-1.5).
- **Methods after close** → (nil, "sqlite: db closed").
- **Wrong argument types** → raises via luaL_error.

Design decisions

- **WAL is opt-in, not default.** WAL is strictly better for most use cases, but it creates *-wal and *-shm sidecar files that some users find surprising. Opt-in keeps the default surprise-free, but every long-running daemon should pass wal=true.
- **Foreign keys ON by default**, contrary to SQLite default. Most modern schemas rely on FK constraints ; defaulting them off is a footgun.
- **transaction(fn) commits if fn returns truthy.** Conventional in Lua : return a value to signal success, return nothing/nil/false (or raise) to rollback. The function's return value is propagated.
- **Placeholders without params is an error**, not a silent NULL bind. Catches a class of injection-adjacent bugs early.
- **Errors are returned, not raised.** Even malformed SQL returns (nil, err). Scripts that don't check are noisy but not catastrophic.

Not in v1

- BLOB streaming I/O (sqlite3_blob_open). Use string serialisation if you fit in memory.
- Virtual tables / FTS5 / R-Tree. Available via raw SQL if the vendored SQLite is built with them ; no Lua-level API yet.
- Backup API (sqlite3_backup_init). For now, VACUUM INTO 'backup.db' works as a one-liner.

For a higher-level ORM-like layer, build it in Lua on top of this API.

[English](#) | [Français](#)

luapilot strings — string manipulation

A single function : split a string by a separator into a table. Predates the sub-namespace convention, lives directly on luapilot.

Why

Splitting a string by a delimiter is one of the most common string operations, and Lua's standard library doesn't have one out of the box (string.gmatch works but requires pattern escaping). A dedicated function is more discoverable and avoids the pattern-escaping trap.

API

Function	Returns
<code>luapilot.split(s, sep)</code>	table (array) of substrings

- `s` : the string to split.
- `sep` : the separator string (literal, no patterns).

If `sep` doesn't appear in `s`, the result is a one-element table containing the whole `s`. If `s` is empty, returns an empty table.

Quick example

```
local parts = luapilot.split("a,b,c,d", ",")
-- parts == { "a", "b", "c", "d" }

local one = luapilot.split("hello", ",")
-- one == { "hello" }
```

Error contract

- **Wrong argument types** → raises via `luaL_error`.
- Otherwise always succeeds.

Design decisions

- **Literal separator, not Lua pattern.** The common case is splitting CSV-like data, where you want `.` to mean a literal dot, not “any character”. If you need pattern matching, fall back to `string.gmatch`.
- **Empty entries are preserved.** `"a,,b"` splits into `{"a", "", "b"}`. Consumers who want to filter empty strings do it in one line of Lua.

Not in v1

- Pattern-based splitting (with escape rules). Use `string.gmatch` if needed.
- `joinTable(t, sep)` mirror — `table.concat` already exists in the stdlib.

[English](#) | [Français](#)

luapilot sys — system utilities

Process and host introspection helpers : PID, hostname, kernel info, executable lookup, environment variables.

Why

Lua's standard library has `os.getenv` and `os.execute`, but nothing for the PID, hostname, kernel info, or a portable `which`. These are universal scripting needs that don't deserve a roundabout via `popen` calls.

These functions live directly on `luapilot` (flat, no sub-table) because they predate the per-module convention used by recent additions.

API

Function	Returns
<code>luapilot.pid()</code>	integer — current process ID
<code>luapilot.hostname()</code>	string (nil, err) — system hostname
<code>luapilot.uname()</code>	table with <code>sysname</code> , <code>nodename</code> , <code>release</code> , <code>version</code> , <code>machine</code>
<code>luapilot.which(cmd)</code>	string (absolute path) (nil, "not found")
<code>luapilot.env(name)</code>	string nil — like <code>os.getenv</code> , but consistent

Function	Returns
<code>luapilot.setenv(name, value)</code>	<code>(true, nil) (nil, err)</code>
<code>luapilot.getMemoryUsage()</code>	<code>integer</code> — Lua VM memory in bytes (after a full GC)
<code>luapilot.getDetailedMemoryUsage()</code>	<code>(integer, integer)</code> — currently both equal the GC count in bytes ; kept as two returns for API stability

Runtime constants

The same `luapilot` table also exposes constants describing the LuaPilot binary that's running the script. Useful for logging, gating features behind a minimum version, or sanity-checking the runtime.

Constant	Type	Example
<code>luapilot.VERSION</code>	<code>string</code>	<code>"1.7.1"</code>
<code>luapilot.VERSION_MAJOR</code>	<code>integer</code>	<code>1</code>
<code>luapilot.VERSION_MINOR</code>	<code>integer</code>	<code>7</code>
<code>luapilot.VERSION_PATCH</code>	<code>integer</code>	<code>1</code>

The string version is what `luapilot --version` prints. The integer components are there for programmatic comparison without parsing the string.

```
-- Log the runtime
print("running under LuaPilot " .. luapilot.VERSION)

-- Gate a feature behind a minimum version
local need_minor = 7
if luapilot.VERSION_MAJOR < 1
  or (luapilot.VERSION_MAJOR == 1 and luapilot.VERSION_MINOR < need_minor) then
  error("this script needs LuaPilot >= 1." .. need_minor)
end
```

Quick example

```
print("Running on:", luapilot.hostname(), "PID:", luapilot.pid())

local u = luapilot.uname()
print("Kernel:", u.sysname, u.release, u.machine)

-- Find a binary
local sh, err = luapilot.which("bash")
if sh then
  luapilot.exec({ sh, "-c", "echo hello" })
end

-- Read/write env
print("HOME:", luapilot.env("HOME"))
luapilot.setenv("MY_VAR", "value")

-- Memory introspection (forces a full GC first)
print("Lua VM memory:", luapilot.getMemoryUsage(), "bytes")
local a, b = luapilot.getDetailedMemoryUsage()
print("Detailed:", a, b)
```

Error contract

- `pid()` : never fails, always returns an integer.

- `hostname()` / `which()` : (value, nil) on success, (nil, err_string) on failure.
- `env(name)` : value if set, nil if unset. Never raises.
- `setenv(name, value)` : (true, nil) on success, (nil, err) on failure (rare — usually OOM or invalid name).
- `uname()` : never fails on any supported system, always returns the full table.
- **Wrong argument types** (e.g. `which(42)`) → raises via `luaL_error` after string coercion as per Lua convention. Pass a table or boolean to force a real error.

Design decisions

- `env(name)` returns **nil**, not "", for unset variables. Matches `os.getenv` semantics. Tests should use `env("X") == nil`, not `env("X") == ""`.
- `which` returns the **absolute path**, not just “yes/no”. This is more useful : you can `exec` the returned path directly. For a boolean check, use `which(cmd) ~= nil`.
- `uname()` returns a **table**, not **multiple values**. Makes it forward-compatible if a field is ever added (e.g. `domainname`).

Not in v1

Additive — could be added later :

- `unsetenv` (just `setenv(name, nil)` could work too).
- `getuid` / `getgid` / `getppid` / `getlogin`.
- Full environment dump as a table.

For user lookups (UID → name, etc.) see the dedicated [user](#) module — that uses NSS, which is the right backend for that question.

[English](#) | [Français](#)

luapilot tables — table manipulation helpers

Two helpers that fill obvious gaps in the Lua standard library : a proper deep copy, and a multi-source merge.

Why

Lua’s standard library has no built-in deep copy and no multi-source table merge. Both are routine needs : merging config from defaults + overrides, snapshotting a table before mutation, copying a tree of options. Writing them per-script is repetitive and error-prone (missing the metatable, cycles, etc.).

These two functions live directly on the `luapilot` table (no sub-namespace) because they predate the per-module convention used by recent additions.

API

Function	Returns
<code>luapilot.mergeTables(t1, t2, ...)</code>	new table — string keys merged (last writer wins), numeric keys appended in order
<code>luapilot.deepCopyTable(t)</code>	new table , recursively copied

`mergeTables` is variadic — pass any number of tables. The inputs are not mutated. The merge has two rules :

- **String keys** : later tables overwrite earlier ones at the same key (last writer wins).
- **Numeric keys (array part)** : concatenated in the order tables are passed, then in 1..n order within each table. So `mergeTables({1, 2}, {3, 4})` gives `{1, 2, 3, 4}`, not `{3, 4}`.

`deepCopyTable` copies nested tables recursively. Cycles are detected (no infinite loop). Non-table values (numbers, strings, booleans, functions, userdata) are not duplicated — they’re referenced as-is.

Quick example

```
-- String keys : last writer wins
local defaults = { port = 8080, host = "localhost", debug = false }
local user_cfg = { port = 9000, debug = true }
local cfg = luapilot.mergeTables(defaults, user_cfg)
-- cfg.port == 9000      (user_cfg override)
-- cfg.host == "localhost"
-- cfg.debug == true

-- Numeric keys : append in order
local a = { 1, 2 }
local b = { 3, 4 }
local merged = luapilot.mergeTables(a, b)
-- merged == { 1, 2, 3, 4 } (NOT { 3, 4 })

-- Deep copy : snapshot before mutation
local snapshot = luapilot.deepCopyTable(cfg)
cfg.port = 9999          -- does NOT affect snapshot
print(snapshot.port)     -- still 9000
```

Error contract

- **Wrong argument types** (passing a non-table to either function) → raises via `luaL_error`.
- Neither function returns errors via `(nil, err)`. They either succeed or raise.

Design decisions

- **Shallow values are referenced, not deep-copied**, in `deepCopyTable`. Copying functions or userdata makes no sense, and copying immutable values (numbers, strings, booleans) changes nothing. Only tables are recursively duplicated.
- **Metatables are not copied** in `deepCopyTable`. This is a conscious choice : preserving a metatable across a copy can surprise callers (the copy still triggers `__index` callbacks that mutate the original). If you need to copy with metatable, do it explicitly with `setmetatable(luapilot.deepCopyTable(t), getmetatable(t))`.
- **mergeTables is shallow**. If two source tables share a sub- table at the same string key, the result references the last one unchanged — it does not recurse into it. Combine with `deepCopyTable` if you need merge-then-isolate semantics.
- **Numeric keys are appended, not overwritten**. The choice comes from the most common Lua use case : combining list-shaped tables (`{1, 2} + {3, 4} → {1, 2, 3, 4}`). If you wanted overwriting semantics on numeric keys, build a destination table and assign explicitly.

Not in v1

Additive — could be added later without breaking SemVer :

- A `deepMergeTables` that recurses into nested tables when both sources have a table at the same key. Common request, but the semantics around lists vs maps gets ambiguous, hence not in v1.
- An `equalsTables` for deep structural comparison. Easy to write in pure Lua, hasn't surfaced as a real need yet.

[English](#) | [Français](#)

luapilot time — monotonic and realtime clocks

Two clocks with sub-second precision and a sleep that honours both `s/ms/us/ns` units and signals. Fills the gaps in `os.time` / `os.clock`. Lives directly on `luapilot`, no sub-namespace.

Why

Lua's `os.time()` only gives seconds, and `os.clock()` measures CPU time, not wall time. Neither is appropriate for measuring real durations (request latency, retry backoff, timeouts). And `os` has no portable nanosecond sleep.

`luapilot.monotonic()` and `luapilot.now()` expose `CLOCK_MONOTONIC` (durations) and `CLOCK_REALTIME` (timestamps), with `luapilot.sleep()` honouring signal interruption.

API

Function	Returns
<code>luapilot.monotonic()</code>	<code>number</code> — seconds since arbitrary epoch (immune to clock changes)
<code>luapilot.now()</code>	<code>number</code> — POSIX time (seconds since 1970-01-01 UTC)
<code>luapilot.sleep(amount, unit?)</code>	<code>(true, nil) (nil, "interrupted")</code>

unit for sleep is one of "s" (default), "ms", "us", "ns". Floats are accepted.

API — date and duration utilities (v1.8.0+)

Added under the `luapilot.time` sub-table. The three flat functions above are also aliased there (`luapilot.time.now`, `luapilot.time.monotonic`, `luapilot.time.sleep`) for ergonomics.

Function	Returns
<code>luapilot.time.iso(ts?)</code>	<code>string</code> — "YYYY-MM-DDTHH:MM:SSZ" (UTC). Without arg, formats the current time.
<code>luapilot.time.parse_iso(s)</code>	<code>(integer, nil) (nil, "parse_iso: ...")</code> — Unix timestamp in seconds.
<code>luapilot.time.parse_duration(s)</code>	<code>(integer, nil) (nil, "parse_duration: ...")</code> — number of seconds.
<code>luapilot.time.format_duration(n)</code>	<code>string</code> — "1d2h3m4s" style, compact, with zero components omitted.

ISO 8601 format accepted by `parse_iso` — pragmatic, but with one strict rule : **a timezone suffix is required**.

- Date and time separator : T or space.
- Timezone : Z (UTC), +HH:MM, or -HH:MM. **No** +0200, +02, or trailing absence — these are all rejected. Hours 00..23, minutes 00..59.
- Fractional seconds (.123) are accepted but ignored.
- A string without timezone is rejected explicitly rather than interpreted as UTC, because most real-world timestamps without a Z are local, not UTC, and silent misinterpretation is a common bug source.

Duration format accepted by `parse_duration` :

- Units : s, m (minute), h, d. No w, mo, y, ms, us, ns in v1.
- Compose unit chunks without whitespace : "1h30m", "2d12h", "45m30s", "1d1h1m1s".
- Units must appear in **strictly decreasing order** (d > h > m > s). "30m1h" is rejected.
- No duplicate unit : "1h1h" is rejected.
- No sign, no spaces. Empty string is rejected. A bare number without unit ("5") is rejected.
- "0s" is the only canonical representation of zero (matches `format_duration(0)`).

Round-trip invariant : for every integer `n >= 0`, `parse_duration(format_duration(n)) == n`. This is checked in the test suite over a representative sample.

Quick example

```
-- Measure a duration safely (immune to NTP / DST jumps)
local t0 = luapilot.monotonic()
do_something()
local elapsed = luapilot.monotonic() - t0
print(string.format("took %.3f s", elapsed))

-- Timestamp an event
local now = luapilot.now()
db:exec("INSERT INTO events (ts, kind) VALUES (?, ?)", { now, "ping" })

-- Sleep 100 ms, but wake up on a signal
local ok, err = luapilot.sleep(100, "ms")
if not ok and err == "interrupted" then
    print("woken by a signal")
end

-- ISO 8601 timestamp for logs (always UTC)
print("event at " .. luapilot.time.iso())          -- "2026-06-17T14:32:01Z"
print(luapilot.time.iso(0))                        -- "1970-01-01T00:00:00Z"

-- Parse a log line back to a Unix timestamp
local ts = assert(luapilot.time.parse_iso("2026-06-17T10:00:00+02:00"))
-- ts is 1750147200 (which is 08:00:00 UTC)

-- Human-readable durations
local cache_ttl = luapilot.time.parse_duration("2h30m") -- 9000
print("cache valid for " .. luapilot.time.format_duration(cache_ttl))
-- "cache valid for 2h30m"
```

Error contract

- **monotonic()** / **now()** : never fail. Always return a number.
- **sleep(amount, unit)** :
 - (true, nil) if the sleep completed normally.
 - (nil, "interrupted") if a handled signal arrived during the sleep (see [signal](#)).
- NaN / Inf / negative amount → raises via `luaL_error`.
- Unknown unit string → raises via `luaL_error`.

Design decisions

- **Two clocks, two functions, no flag.** `monotonic()` for durations, `now()` for timestamps. Conflating them in one function with a parameter would invite the wrong choice.
- **now()** for realtime, not `realtime()` — shorter, and the natural English meaning matches what’s expected (current time).
- **Signal-aware sleep.** A 60-second sleep that ignores `SIGTERM` is the classic graceful-shutdown bug. `sleep()` returns (nil, "interrupted") so the caller can exit cleanly.
- **No “format date” helper.** `os.date(format, luapilot.now())` works fine, and a duplicate function would just be a wrapper.

Not in v1

- `luapilot.boottime()` (`CLOCK_BOOTTIME` — includes suspend). Rarely needed, easy to add later.
- `luapilot.utcnw()` / `localnow()` helpers that return pre-formatted strings. Trivial in Lua, no value-add.

[English](#) | [Français](#)

luapilot.socket TLS — connect_tls and starttls

The TLS half of `socket` : encrypted TCP for IRC over TLS, IMAPS, custom secure protocols. Two entry points : `connect_tls` for protocols that go TLS from the start (port 6697 IRC, 993 IMAPS, etc.) and `starttls` for protocols that upgrade an existing plain connection (SMTP, IMAP, IRC STARTTLS).

Why

Talking encrypted TCP without `http` is a real need : IRC bots over +6697, SMTP submission, custom internal services behind TLS. Doing it manually with `openssl s_client` via `exec` is unreliable ; `openssl` the C++ library + `cpp-httplib` chosen abstractions give a clean Lua API that handles cert verification correctly by default.

API

Function	Returns
<code>luapilot.socket.connect_tls(host, port, opts?)</code>	<code>tls_socket</code> <code>(nil, err)</code>
<code>s:starttls(opts?)</code>	<code>(true, nil)</code> <code>(nil, err)</code> — upgrade an existing plain socket

`opts` (merged with the usual socket `opts`) :

Field	Type	Default
<code>verify</code>	boolean	<code>true</code>
<code>ca_cert</code>	string (path)	system bundle
<code>ca_path</code>	string (path)	system bundle
<code>server_name</code>	string	host argument
<code>alpn</code>	array of strings	<code>none</code>

A `tls_socket` has the same methods as a plain socket (`send`, `recv`, `recv_line`, `recv_all`, `set_timeout`, `close`, `peer`). The encryption is transparent.

Quick examples

Connect to IRC over TLS

```
local s = assert(luapilot.socket.connect_tls("irc.libera.chat", 6697, {
    timeout = 30,
}))
```

```
s:send("NICK luapilot-bot\r\n")
s:send("USER luapilot 0 * :luapilot bot\r\n")
```

```
while true do
    local line, err = s:recv_line()
    if not line then break end
    print(line)
    if line:match("^PING (.+)$") then
        s:send("PONG " .. line:match("^PING (.+)$") .. "\r\n")
    end
end
```

STARTTLS upgrade (SMTP submission)

```
local s = assert(luapilot.socket.connect("mail.example.com", 587))
print(s:recv_line())           -- 220 banner
```

```

s:send("EHLO myhost\r\n")
repeat
    line = s:recv_line()
until not line:match("^250%-")           -- last 250 line

s:send("STARTTLS\r\n")
print(s:recv_line())                    -- 220 ready to start TLS

assert(s:starttls({ server_name = "mail.example.com" }))
-- s is now encrypted ; continue with EHLO + AUTH + ...

```

Self-signed server (dev / private CA)

```

local s = assert(luapilot.socket.connect_tls("internal.svc", 5555, {
    ca_cert = "/etc/myapp/internal-ca.crt",
}))

```

Skip verification (TESTING ONLY)

```

local s = assert(luapilot.socket.connect_tls("self-signed.local", 8443, {
    verify = false,
}))

```

Error contract

- All errors prefixed "tls: ..." (handshake, cert verify, etc.) or "socket: ..." (DNS, connect, timeout).
- (nil, err) always — no exceptions for “expected” failures like cert mismatch.
- **verify=true + invalid cert** → (nil, "tls: certificate verify failed: <reason>"). Common reasons : expired, self-signed, hostname mismatch, unknown CA.
- **NaN/Inf timeouts**, wrong types → raises via `luaL_error`.

Trust store

LuaPilot ships its own statically-linked OpenSSL. Out of the box, `verify=true` works on :

- **Arch, Debian, Ubuntu, Alpine, Gentoo** : via `--openssl-dir=/etc/ssl` baked into the vendored OpenSSL.
- **Fedora, RHEL, CentOS, Rocky, Alma, OpenSUSE, FreeBSD, NetBSD** via runtime probing of known CA bundle paths.

Overrides, in priority order :

1. `ca_cert` / `ca_path` per call.
2. `SSL_CERT_FILE` / `SSL_CERT_DIR` environment variables.
3. The two mechanisms above.

If none of these find a CA store and you use `verify=true`, the handshake fails with a clear error. See [security](#) for the full picture.

Design decisions

- **TLS 1.2 minimum**. SSL 3, TLS 1.0, TLS 1.1 are unconditionally rejected — they’re broken, no modern server should rely on them. TLS 1.3 is negotiated automatically when available.
- **verify=true by default, verify=false requires explicit opt-in**. No way to accidentally turn off cert checking.
- **server_name defaults to the host argument** to make SNI Just Work. Pass it explicitly if you’re connecting by IP to a TLS server that uses SNI.
- **Same methods as plain socket**, so a function that takes a `socket` works for both transparently.

Not in v1

- Client certificate authentication. Possible to add as a `client_cert` / `client_key` opt later.
- TLS session resumption / session tickets. Not commonly needed at script level.
- OCSP stapling verification. Reliance on OpenSSL's defaults.

[English](#) | [Français](#)

luapilot.toml — TOML parser

Backed by [toml++](#), a strict TOML 1.0 parser. Decode only — encoding TOML faithfully from a Lua table is ambiguous (Lua doesn't distinguish between inline tables and dotted tables, for instance).

Why

TOML is the natural config-file format for modern tools (Cargo, pyproject, systemd-credentials, etc.). A LuaPilot script reading its own config should be able to do it directly, not via a TOML → JSON conversion via `tomlq`.

API

Function	Returns
<code>luapilot.toml.decode(text)</code>	<code>table</code> (parsed TOML) <code>(nil, err)</code>

The returned table mirrors the TOML structure :

- Strings → Lua strings (UTF-8)
- Integers → Lua integers
- Floats → Lua numbers
- Booleans → Lua booleans
- Datetimes → Lua strings in ISO 8601 form (TOML has no Lua-native equivalent ; use `os.date` or a date library if you need parsed components)
- Arrays → Lua tables (1-indexed)
- Tables (`[section]` or `{ inline }`) → Lua tables (with string keys)

Quick example

```
local body = [[
title = "My App"
[server]
host = "localhost"
port = 8080
features = ["auth", "metrics"]

[database]
url = "sqlite:///var/lib/app.db"
]]

local cfg, err = luapilot.toml.decode(body)
if not cfg then error("config parse failed: " .. err) end

print(cfg.title)           -- "My App"
print(cfg.server.host)     -- "localhost"
print(cfg.server.features[1]) -- "auth"
print(cfg.database.url)    -- "sqlite:///var/lib/app.db"
```


Error contract

- **decode** : (nil, err) on parse error. The error message includes the line/column where parsing failed, and a short description of what was wrong (per toml++).
- **Non-string argument** → raises via `luaL_error`.

Design decisions

- **Decode only, no encode**. Round-tripping a TOML file via a Lua table loses formatting (comments, dotted vs inline tables, newlines between sections) that real users care about. If you need to write TOML, build the string yourself — TOML is designed to be human-writable. A faithful encoder would be a large undertaking for limited value.
- **Datetimes as strings**. TOML's datetime type is rich (offset, local, date-only, time-only) but Lua has no native equivalent. Returning a string in ISO 8601 form preserves the data losslessly and lets the script parse it with the tools it prefers.

Not in v1

- Encoder. May be added if a real use case appears, but unlikely.
- A schema validator. Build one in Lua if you need it ; the decoded structure is just a table.

[English](#) | [Français](#)

luapilot.user

System user lookups via NSS, without parsing `/etc/passwd` directly.

Why

On a modern Linux system, users may come from multiple sources : `/etc/passwd`, LDAP, SSSD, NIS+, FreeIPA, systemd-userdb, or other NSS plugins. Reading `/etc/passwd` directly misses everything except the first source. `luapilot.user` is backed by `getpwnam_r(3)` and `getpwuid_r(3)`, which traverse the resolver chain configured in `/etc/nsswitch.conf` — exactly what `id` or `getent passwd` does.

API

Function	Returns
<code>user.get(name_or_uid)</code>	<code>table</code> <code>(nil, "user not found")</code> <code>(nil, "user: <err>")</code>
<code>user.exists(name_or_uid)</code>	<code>boolean</code>

`name_or_uid` accepts :

- a **string** → looked up by name via `getpwnam_r`
- a **non-negative integer** → looked up by UID via `getpwuid_r`

Any other type, a float, a negative integer, a string containing an embedded NUL byte, or an integer above `uid_t` max raise via `luaL_error` — these are caller bugs, not runtime conditions to handle gracefully.

Returned table on success

```
{
  name  = "yaourt",
  uid   = 968,
  gid   = 968,
  gecos = "yaourt AUR build user",
  home  = "/var/cache/yaourt",
  shell = "/usr/sbin/nologin",
}
```

All string fields are guaranteed non-nil. They may be empty strings when the underlying NSS source has no value (this is rare but possible — typically empty `gecos`).

Quick example

```
-- Ensure a service user is provisioned before launching the daemon.
if not luapilot.user.exists("yaourt") then
    error("yaourt user not provisioned — run useradd first")
end

local u = assert(luapilot.user.get("yaourt"))
luapilot.chdir(u.home)
```

Error contract

- **Bad argument type** (table, boolean, nil, float, negative integer, UID above `uid_t` max, string with embedded NUL) → raises via `luaL_error`. Use `pcall` if you need to recover.
- **User absent** (the resolver returned no match) → `get()` returns (nil, "user not found") ; `exists()` returns false.
- **NSS error** (resolver itself broken : LDAP unreachable, out of memory, etc.) → `get()` returns (nil, "user: <description>") ; `exists()` returns false (errors are silently coerced into “absent” for predicates ; use `get()` if you need to diagnose).

Design decisions

- **NSS only, never `/etc/passwd`** : reading the file directly would silently miss every account that isn’t in the local file. The whole point of `luapilot.user` is that the script gets the same answer as `id` or `getent passwd`.
- **Thread-safe `_r` variants** : `LuaPilot` exposes `luapilot.workers`, so we use `getpwnam_r/getpwuid_r` everywhere.
- **`gecos` exposed raw** : the historical format is `Full Name,Office,WorkPhone,HomePhone[,Other]`, but in practice most entries hold a free-form name. Parsing it in Lua is trivial if you need to ; baking a parser in C++ would be premature.
- **NUL byte rejection** : `"root\0evil"` would be seen as `"root"` by `getpwnam_r` (it stops at the first NUL). On user-derived input, that could bypass an upstream identity check. Explicit rejection is safer than implicit truncation.
- **`uid_t` range check** : on Linux, `uid_t` is `uint32_t`, while `lua_Integer` is `int64_t`. Without a check, `5_000_000_000` would be silently truncated to a 32-bit value and could match an unrelated account.

Not in v1

Additive — these could be added later without breaking SemVer :

- Group lookups (`getgrnam` / `getgrgid`) — will be added when a concrete use case asks for them.
- `/etc/shadow` access (passwords, password aging) — out of scope. Requires root and is rarely useful outside niche admin scripts. Password authentication should go through PAM, not `LuaPilot`.
- User/group creation — already feasible via `luapilot.exec("useradd ...")` and `groupadd`. No need for a dedicated binding.

[English](#) | [Français](#)

`luapilot.workers` — parallel jobs

Run Lua code on real OS threads. Useful for I/O-bound work that benefits from concurrency (fetching many URLs, scanning many files, processing batches) and CPU-bound work on multi-core hardware.

Why

Lua has coroutines, but those run cooperatively on a single OS thread — fine for state machines, not for using multiple cores. And many tasks are I/O-bound : 100 HTTP requests that each take 500 ms still take 50 s sequentially,

but ~500 ms in parallel.

`luapilot.workers` spawns real OS threads each running their own isolated Lua state, communicating with the parent via passed-in arguments and return values. No shared state means no locks, no data races — just send work in, get results out.

API

Function	Returns
<code>luapilot.workers.spawn(code, args?)</code>	<code>job (userdata) (nil, err)</code>
<code>luapilot.workers.cpu_count()</code>	<code>integer</code> — number of logical CPUs
<code>job:join(timeout?)</code>	<code>(true, result) (false, err) (nil, "timeout")</code>
<code>job:done()</code>	<code>boolean</code> — non-blocking check
<code>job:cancel()</code>	<code>(true, nil)</code> — cooperative ; sets a flag

code argument

A Lua source string that runs in the worker. The full `luapilot` namespace is available, plus a `worker` global with :

- `worker.args` — the second argument passed to `spawn`.
- `worker.cancelled()` — returns `true` if the parent called `job:cancel()`. The worker is expected to check this in long loops.

The worker's last expression value (or return value) is what `join()` returns as `result`.

args argument

Any value that can be deep-copied between Lua states : `nil`, `booleans`, `numbers`, `strings`, `tables` of the same. **Not** : `functions`, `userdata`, `threads`, `tables with cycles`. `Tables` are deep-copied — no sharing.

Quick examples

Parallel HTTP fetches

```
local urls = {
  "https://a.example/",
  "https://b.example/",
  "https://c.example/",
  "https://d.example/",
}

local jobs = {}
for i, url in ipairs(urls) do
  jobs[i] = luapilot.workers.spawn([[
    local url = worker.args.url
    local r, err = luapilot.http.get(url, { timeout = 10 })
    if not r then return { ok = false, err = err } end
    return { ok = true, status = r.status, length = #r.body }
  ]], { url = url })
end

for i, job in ipairs(jobs) do
  local ok, result = job:join()
  if ok then
    print(i, result.status or result.err)
  else
    print(i, "worker crashed:", result)
  end
end
```

```

end
end

```

CPU-bound work with cancellation

```

local n = luapilot.workers.cpu_count()
print("running on", n, "cores")

local jobs = {}
for i = 1, n do
    jobs[i] = luapilot.workers.spawn([[
        local chunk = worker.args.chunk
        local total = 0
        for x = chunk.from, chunk.to do
            if x % 1000 == 0 and worker.cancelled() then
                return { cancelled = true, partial = total }
            end
            total = total + heavy_compute(x)
        end
        return { total = total }
    ]], { chunk = { from = i * 1000, to = (i + 1) * 1000 - 1 } })
end

-- ... later, if needed:
-- for _, j in ipairs(jobs) do j:cancel() end

```

Worker pool pattern

```

local function map_parallel(items, code, max_concurrent)
    max_concurrent = max_concurrent or luapilot.workers.cpu_count()
    local results = {}
    local i = 1
    local active = {}

    while i <= #items or next(active) do
        -- launch as many as we can
        while i <= #items and #active < max_concurrent do
            active[#active + 1] = {
                index = i,
                job = luapilot.workers.spawn(code, { item = items[i] }),
            }
            i = i + 1
        end
        -- harvest any finished
        for k = #active, 1, -1 do
            local entry = active[k]
            if entry.job:done() then
                local ok, result = entry.job:join()
                results[entry.index] = ok and result or { err = result }
                table.remove(active, k)
            end
        end
        if next(active) then luapilot.sleep(10, "ms") end
    end
    return results
end

```

Error contract

- **spawn** : (nil, err) if the OS thread can't be created (rare — usually resource limits).
- **join** :
 - (true, result) — worker completed normally ; **result** is its return value (or nil if it didn't return).
 - (false, err) — worker crashed with an uncaught error ; **err** is the error message + traceback.
 - (nil, "timeout") — timeout elapsed without the worker finishing.
- **cancel** never fails. The flag is set ; the worker may take a while to notice (it must call `worker.cancelled()`).
- **Wrong argument types** → raises via `luaL_error`.

Sharing data

Workers don't share Lua state. Communication is :

- **In** : via the `args` second argument to `spawn` (deep-copied into the worker's state).
- **Out** : via the worker's return value (deep-copied back).
- **Side effects** : a worker can write to a file, append to a log, query a database — but coordination through such side effects is the script's responsibility.

If two workers write to the same SQLite file, set `wal=true` on `open` so they don't lock each other out. SQLite handles the synchronisation correctly with WAL ; in `busy_timeout` you can configure how long to wait on a busy db.

Design decisions

- **One Lua state per worker, no sharing.** The alternative — shared state with locks — is a well-known source of subtle bugs and we don't think Lua-level locks are worth the design effort at this layer. Pure message-passing is simpler.
- **No global thread pool.** Each `spawn` creates a fresh OS thread, each `join` closes it. For tight loops, build a pool in Lua (see example above). The simpler model wins on legibility.
- **Cooperative cancellation, not preemptive.** There's no way to forcibly stop a Lua thread in the middle of an operation without leaving the runtime in an undefined state. `worker.cancelled()` returns `true` ; the worker is responsible for checking it periodically.
- **luapilot.signal is not available in workers.** Signals are process-wide ; only the main thread can sensibly own them.

Not in v1

- Channels (FIFO queues between workers). Add later if a use case shows the pattern is common.
- Shared memory (mmap). Same.
- Async I/O futures (`spawn().then(...)` style). Out of scope ; use a polling loop or `done()` checks.

[English](#) | [Français](#)

Cookbook

Concrete recipes for common scripting tasks. Each recipe is short and self-contained ; copy, adapt, ship.

Watch a directory and email new files

A drop directory that mails out every file as it arrives. Uses `luapilot.inotify` for instant wakeup (no polling), and listens on `close_write` + `moved_to` so the file is guaranteed complete.

```
local w = assert(luapilot.inotify.new())
assert(w:add("/srv/incoming", { "close_write", "moved_to" }))

luapilot.signal.handle("TERM", function()
    w:close()
    os.exit(0)
end)
```

```

while true do
    local events, err = w:read()
    if not events then
        if err == "interrupted" then break end
        io.stderr:write("inotify: ", err, "\n"); break
    end
    for _, ev in ipairs(events) do
        if ev.events.overflow then
            -- queue overflowed: rescan the directory
            -- (events were lost)
        else
            local path = "/srv/incoming/" .. ev.name
            send_email(path)
            os.remove(path)
        end
    end
end
end

```

Ensure a service user exists, then drop to it

A typical install-time check before launching a daemon as a non-root user.

```

local function ensure_user(name)
    if luapilot.user.exists(name) then return true end
    local ok, err = luapilot.exec("useradd", {
        "--system",
        "--home-dir", "/var/lib/" .. name,
        "--create-home",
        "--shell", "/usr/sbin/nologin",
        name,
    })
    if not ok then
        return nil, "useradd failed: " .. tostring(err)
    end
    return true
end

assert(ensure_user("myapp"))
local u = assert(luapilot.user.get("myapp"))
luapilot.chdir(u.home)

```

Fetch many URLs in parallel

luapilot.workers runs Lua on multiple cores. Total wall-clock is close to the slowest single fetch, not the sum.

```

local urls = { "https://a.example/", "https://b.example/",
    "https://c.example/", "https://d.example/" }

local jobs = {}
for i, url in ipairs(urls) do
    jobs[i] = luapilot.workers.spawn([[
        local url = worker.args.url
        local r, err = luapilot.http.get(url, { timeout = 10 })
        if not r then return { error = err } end
        return { status = r.status, length = #r.body }
    ]], { url = url })
end

```

```

for i, job in ipairs(jobs) do
    local ok, result = job:join()
    print(i, ok, result and result.status or result.error)
end

```

Graceful shutdown of a long-running script

Catch SIGTERM / SIGINT so the script can close resources properly. Works with `systemctl stop` and Ctrl-C.

```

local log = require("logging")
log.set_level("info")

local running = true
luapilot.signal.handle("TERM", function()
    log.info("SIGTERM, shutting down")
    running = false
end)
luapilot.signal.handle("INT", function()
    log.info("Ctrl-C, shutting down")
    running = false
end)

while running do
    -- main loop ; blocking calls return (nil, "interrupted") when
    -- a handled signal arrives, so the loop exits quickly.
    local line, err = sock:recv_line()
    if not line then
        if err == "interrupted" then break end
        log.error("recv: ", err); break
    end
    process(line)
end

sock:close()
log.info("clean shutdown")

```

Read a TOML config, validate, persist to SQLite

```

-- Load TOML
local f = assert(io.open("config.toml", "r"))
local body = f:read("a"); f:close()
local cfg, perr = luapilot.toml.decode(body)
if not cfg then error("config.toml: " .. perr) end

-- Validate (cheap manual checks ; for a real schema, build one in Lua)
assert(type(cfg.server) == "table", "missing [server]")
assert(type(cfg.server.host) == "string", "missing server.host")
assert(math.type(cfg.server.port) == "integer", "server.port must be integer")

-- Open db, persist
local db = assert(luapilot.sqlite.open("state.db", { wal = true }))
db:exec([[CREATE TABLE IF NOT EXISTS config (k TEXT PRIMARY KEY, v TEXT)])
db:exec("INSERT OR REPLACE INTO config VALUES (?, ?)",
    { "host", cfg.server.host })
db:exec("INSERT OR REPLACE INTO config VALUES (?, ?)",
    { "port", tostring(cfg.server.port) })

```

```
db:close()
```

Cache with a human-readable TTL, expiry as ISO timestamps

Parse a TTL from config ("15m", "2h30m", "1d"), compute a deadline, and log it in a uniform UTC format that grep'ing across log files actually works on.

```
-- Config from TOML, CLI, env, anywhere.
local raw_ttl = "15m"

local ttl, err = luapilot.time.parse_duration(raw_ttl)
if not ttl then error("bad TTL '" .. raw_ttl .. "': " .. err) end

local deadline = luapilot.now() + ttl
print(string.format("[%s] cache entry created, expires at %s (TTL %s)",
    luapilot.time.iso(),
    luapilot.time.iso(deadline),
    luapilot.time.format_duration(ttl)))

-- ...later, somewhere else in the script:
if luapilot.now() >= deadline then
    print "[" .. luapilot.time.iso() .. "] cache miss: entry expired"
end
```

Useful properties of this pattern :

- `luapilot.time.iso()` is always UTC, always 20 characters, always the same shape. `grep` and `sort` work on the timestamps directly.
- `parse_duration("15m")` returns an integer, so `now() + ttl` is exact (no float drift over long durations).
- `format_duration(ttl)` round-trips : `parse_duration( format_duration(n)) == n` for every `n >= 0`. Safe to use as a canonical representation in logs or databases.

Pretty-print a Lua value

The bundled `inspect` module gives readable output for nested tables.

```
local inspect = require("inspect")
print(inspect({ a = 1, b = { c = 2, d = { 3, 4, 5 } } }))
-- { a = 1, b = { c = 2, d = { 3, 4, 5 } } }
```

Hello there

The minimal LuaPilot script :

```
luapilot.helloThere()
```